ITAI 1378 MIDTERM · TIER 2 – CUSTOM DATASET + TRANSFER LEARNING

G.O.A.T Vision: *Game Outcome Action Tracker*

A deep learning computer vision system that automatically classifies soccer play events from images — built by **Samira Gevara** using transfer learning on a custom dataset.

The Problem

What's Happening

Identifying soccer play events from images or match footage is **typically done manually** — a slow, labor-intensive process that doesn't scale with modern data volumes.

Who Cares

- Coaches and analysts reviewing match footage
- Sports broadcasters building highlight reels
- Sports analytics companies processing game data

Why It Matters

Manual event tagging is time-consuming, error-prone, and bottlenecks the entire analytics pipeline. As match footage volumes grow exponentially, the gap between data availability and actionable insight widens.

Automatically identifying key moments — free kicks, penalty kicks, corner kicks, shots on goal, and yellow cards — can dramatically improve the efficiency of **highlight creation, scouting workflows, and real-time match analysis**.

Solution Overview

G.O.A.T Vision is a computer vision model that classifies soccer event images into five distinct play categories using a fine-tuned ResNet50 neural network. Images are preprocessed and passed through the network to produce a predicted event class in real time.

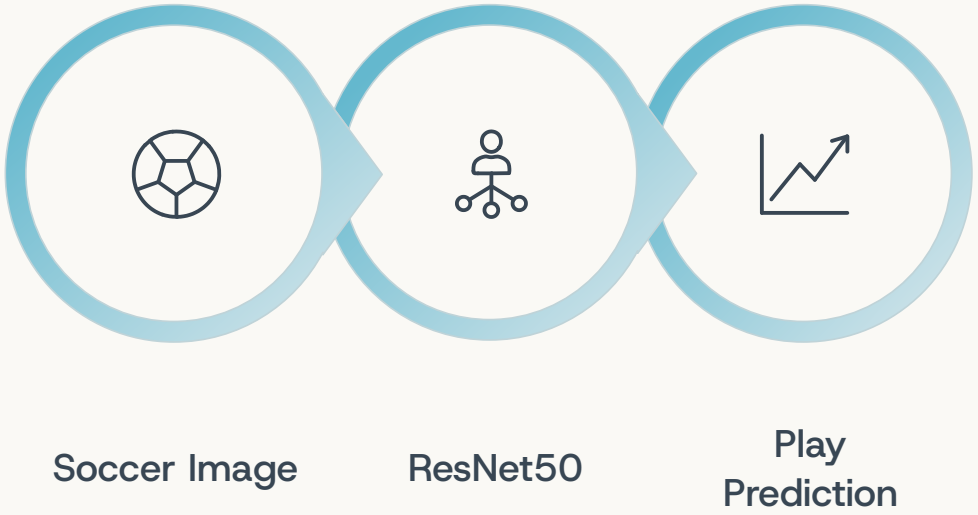
⚽ Corner Kick
Ball delivered from the corner arc into the penalty area

🦵 Free Kick
Set piece from a stoppage outside the box

🎯 Penalty Kick
One-on-one attempt from the penalty spot

🥅 Shot on Goal
Open-play attempt directed at the goal

🟨 Yellow Card
Referee issuing a caution to a player



Technical Approach

Instead of training from scratch, G.O.A.T Vision **fine-tunes a pretrained ResNet50 model** on a custom soccer dataset. Transfer learning lets the model leverage rich visual features already learned from ImageNet, then adapt them to soccer-specific patterns — enabling strong performance even with a small labeled dataset.

Model Architecture

- **Task:** Multi-class image classification
- **Model:** ResNet50 with transfer learning
- **Head:** Custom fully connected classifier (5 output classes)
- **Training:** Google Colab GPU — fast iteration cycles

Frameworks & Tools

- **PyTorch** — model definition and training loop
- **torchvision** — pretrained ResNet50, image transforms
- **scikit-learn** — metrics and confusion matrix
- **matplotlib** — training curves and visualizations
- **Google Colab** — cloud GPU environment

Data Plan

1

Dataset Source

Images manually collected and curated from public soccer imagery sources, ensuring diverse game contexts and visual variety across event types.

2

Dataset Size

~146 labeled images distributed across five event classes. The compact size makes transfer learning the critical enabler for achieving strong accuracy.

3

Data Preparation

Pipeline includes **dataset cleaning**, removal of ambiguous or irrelevant images, and **data augmentation** during training (flips, crops, color jitter) to improve generalization.

4

Class Labels

`corner_kick` · `free_kick` · `penalty_kick` · `shot` · `yellow_card`

AI Pipeline Architecture

G.O.A.T Vision takes a raw soccer image through a structured preprocessing and inference pipeline — from pixel normalization to final event classification — powered by a ResNet50 CNN with transfer learning at its core.

82.14%

Validation Accuracy

Exceeds the 70% target threshold set at project inception

5

Event Classes

Distinct soccer play types classified by the model

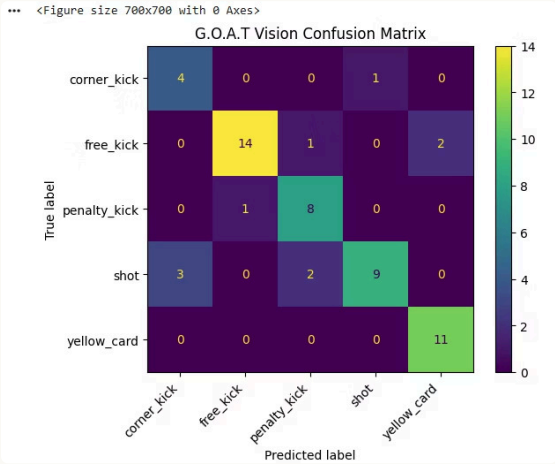
146

Training Images

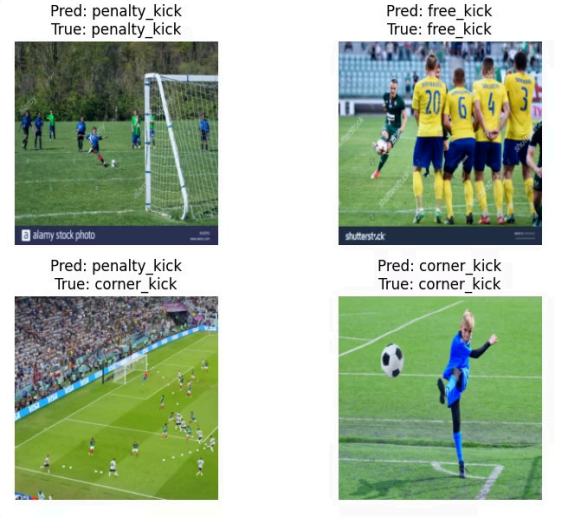
Custom-labeled images used to fine-tune the pretrained model

Success Metrics & Results

Confusion Matrix — G.O.A.T Vision



Sample Predictions



Performance Summary

Target: ≥ 70% validation accuracy
Result: 82.14% — surpassed by +12 points

Key Observations

Yellow card: 11/11 correct — visually distinctive, easiest class

Free kick: 14/17 — strong performance on set-piece imagery

Main confusion: `shot` vs `corner_kick` / `penalty_kick` — similar goal-area camera angles make these the hardest to distinguish

Week by Week Plan

The project follows a structured 6-week execution plan from dataset collection through final presentation, with clear milestones at every stage to ensure steady progress and accountability.

Week	Task	Milestone
Week 11	Dataset collection and GitHub repository setup	Dataset ready
Week 12	Model architecture implementation and initial configuration	Model running
Week 13	Training loop, hyperparameter tuning, and testing	Accuracy achieved
Week 14	Evaluation metrics, confusion matrix, and visualization	Results ready
Week 15	Documentation, README, and slide preparation	Presentation ready
Week 16	**Final presentation — May 8th**	**Project complete**

Challenges & Backup Plans

Two primary risk factors were identified early and addressed with concrete mitigation strategies — ensuring the project could maintain momentum even under adverse conditions.

Challenge 1: Small Dataset Size

With only ~146 labeled images, a model trained from scratch would likely overfit and generalize poorly. Small datasets are a common bottleneck in custom computer vision projects.

Mitigation

Use **transfer learning** with a pretrained ResNet50 to leverage existing visual representations, combined with **data augmentation** (flips, crops, jitter) to artificially expand the effective training set and improve robustness.

Challenge 2: Similar Visual Patterns Between Classes

Several event types — particularly `shot`, `penalty_kick`, and `corner_kick` — share very similar camera angles, field positioning, and player arrangements, making visual discrimination difficult even for human annotators.

Plan B

If per-class accuracy remains unstable, **simplify the classification taxonomy** by merging visually similar categories (e.g., grouping goal-area events) to achieve a more reliable and actionable model output.

Resources Needed

Compute & Infrastructure

- **Google Colab (GPU)** — free cloud GPU for training
- No dedicated hardware required
- **Estimated cost: $0**

Core Frameworks

- **PyTorch** — model training and inference
- **torchvision** — pretrained models and transforms
- **scikit-learn** — evaluation metrics
- **matplotlib** — result visualization

GitHub Repository

All source code, training scripts, dataset structure, and documentation are publicly available on GitHub.

Repository: samirag2010 / ITAI-1378-Midterm-GOATVision-PlayClassifier
G.O.A.T Vision – Soccer Play Type Classifier using PyTorch

Summary

The entire project is built on **open-source tooling at zero cost**, demonstrating that high-accuracy computer vision models are accessible without expensive infrastructure — just transfer learning, a labeled dataset, and a free GPU.